Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Two Reasons

1. A single kiosk with a fixed flow (select coffee, choose size, pay, print receipt) maps naturally onto Presentation / Business / Data layers, keeping the touch UI separate from order and payment logic.

2. Microservices would be operationally heavy and Client-Server would introduce a single point of failure for a standalone device; layers can be deployed together on the kiosk and each layer (e.g. the printer or payment component) can be replaced behind its interface.

Security Advantage

Only the Order Manager talks to the Payment Service, so card data is confined to one business-layer component behind a narrow interface; the UI layer never touches card data or the database directly.

Performance Benefit

All components run in-process on the kiosk, so layer calls are local function calls with no network hops, which keeps a full order (selection to payment) comfortably under the 60-second requirement.